

# Theory of Computation:-

Finite Automata:- A machine having fixed number of states.

\* Types: DFA, NFA, E-NFA

DFA - Deterministic finite Automata

NFA - Non-deterministic

E-NFA - Epsilon of Non-deterministic

Concepts of FA:- \* Alphabet \* String \* Language

\* Alphabets: An Alphabet is finite non-empty set of symbol, denoted by ' $\Sigma$ '.

For example:  $\Sigma = a, \dots, z$ ,  $\Sigma = 0, \dots, 9$ ,  $\Sigma = \text{ASCII} (., ,, \$)$

\* String: String is finite sequence of symbol chosen from some alphabets.

Example: 01101, abab, a12

$\hookrightarrow \Sigma = 0, 1$   $\Sigma = a, b$   $\Sigma = a, \Sigma = 1, 2$

2.1 Empty String: It is a zero occurrence of symbol. denoted by ' $\epsilon$ '

2.2 Length of String: It is total number of Alphabet present in the given string

Ex:  $|01101| = 5$

2.3 Powers of Alphabets: If ' $\Sigma$ ' is a alphabet, we can express the a string of a certain length from that alphabet by using an exponential notation. We define ' $\Sigma^k$ ' to be the set of string of length  $k$  each of whose symbol is ' $\Sigma$ '

$$\Sigma = \{a, b, c\}$$

$$\Sigma^1 = \{a, b, c\}$$

$$\Sigma^2 = \{aa, ab, ac, bb, ba, bc, ca, cb, cc\}$$

## 2.4 Concatenation of string:

Ex:  $Y = ab$   $Z = ba$   $YZ = abba$

\* Language: A set of string of all which are chosen from some  $\Sigma^*$  is called a Language

### Notation for Designing a FA:

→ - start state, Move one state to another

○ - State

$q$  - State Name  $\{q_0, q_1, q_2\}$  (2)

⊙ - Final state / Accepting state

Note: FA can only have one start state

\* Can have more than one final state

\* Start state can be act as final state

ex:  $\rightarrow q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \rightarrow \odot$

### Component / Tuple of FA:

$$M = (Q, \Sigma, \delta, q_0, F)$$

$Q$  → state, Finite no of states

$\Sigma$  - Alphabet

$\delta$  - Transition Function / Move

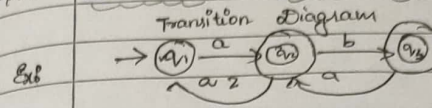
$\odot \rightarrow \delta(q_0, a) = q_1$  N.S

$q_0$  - start state

$F$  - Set of F.S

## Deterministic Finite Automata (DFA):

Transition Diagram



$$M = (Q, \Sigma, \delta, q_0, F)$$

(i)  $Q = \{q_1, q_2, q_3\}$

(ii)  $\Sigma = \{a, b\}$

(iii)  $\delta(q_1, a) = q_2$  ,  $\delta(q_2, b) = q_3$

$\delta(q_3, a) = q_1$  ,  $\delta(q_2, a) = q_1$

(iv)  $q_0 = q_1$

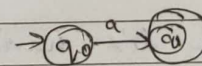
(v)  $F = \{q_3, q_2\}$

### Transition Table

| $Q \backslash E$ | a     | b      |
|------------------|-------|--------|
| $q_1$            | $q_2$ | $\phi$ |
| $q_2$            | $q_1$ | $q_2$  |
| $q_3$            | $q_2$ | $\phi$ |

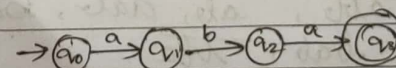
Q: Design a DFA which accept language 'a'

$$M = (Q, \Sigma, \delta, q_0, F)$$



Q: Design a DFA which accept language 'ab'

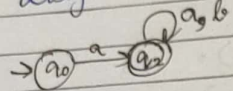
$$M = (Q, \Sigma, \delta, q_0, F)$$



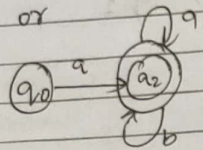
Q Design DFS which accept a empty ( $\epsilon$ ) string



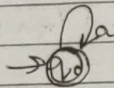
Q Design DFS which start with a 'a',  $\Sigma = a, b$   
 $S = a, b$   $S = a, aa, aaa, \dots$



$S = a, aa, ab, aaa, aab, aba$

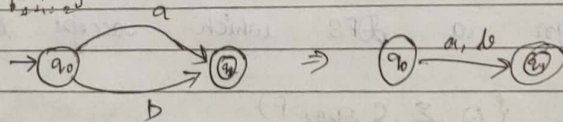


Q Design DFS for any numbers of state for a



Q Design DFS which (i) accept one a or one b

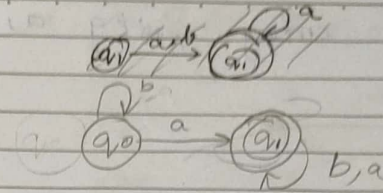
$\Sigma = \{a, b\}$



(ii) accept string of a's and b's having atleast one a

$\Sigma = a, b$   
 Possible  $S = a, ab, ab, aab, baa, bab, bba$

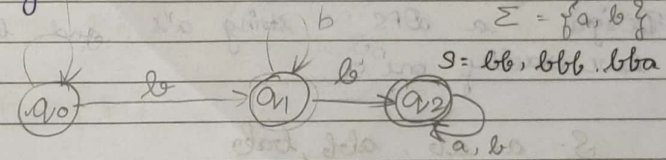
$S = a, ba, bba, baa, aaa, \dots$



Transition Table

| Q/E | a  | b  |
|-----|----|----|
| q0  | q1 | q0 |
| q1  | q1 | q1 |

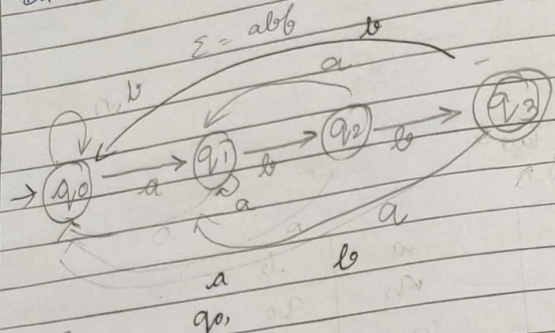
Design DFS to accept string a's & b's starting 'bb'



Transition state

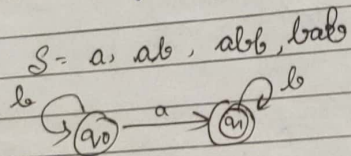
| Q/E | a  | b  |
|-----|----|----|
| q0  |    | q1 |
| q1  |    | q2 |
| q2  | q2 | q2 |

Draw a DFS ending string a b b



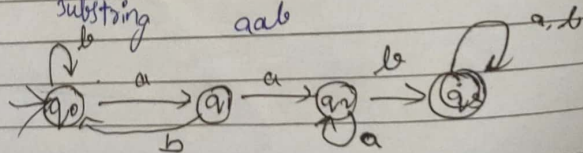
q0  
q1  
q2  
q3\*

Design a DFA string a's and b's have exactly one a



Trap State:-

a) DFA to accept the string a, b having substring aab



$M = (Q, E, \delta, q_0, F)$

|     | a  | b  |
|-----|----|----|
| q0  | q1 | q0 |
| q1  | q2 | q0 |
| q2  | q2 | q3 |
| q3* | q3 | q3 |

$Q = \{q_0, q_1, q_2, q_3\}$

$E = \{a, b\}$

$\delta(q_0, a) = q_1$      $\delta(q_0, b) = q_0$

$\delta(q_1, a) = q_2$      $\delta(q_1, b) = q_0$

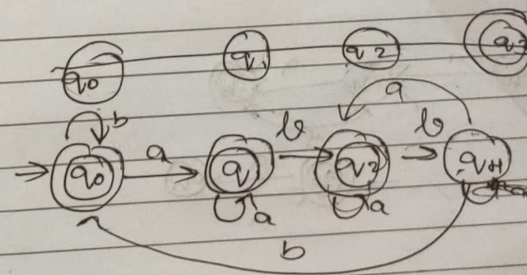
$\delta(q_2, a) = q_2$      $\delta(q_2, b) = q_3$

$\delta(q_3, a) = q_3$      $\delta(q_3, b) = q_3$

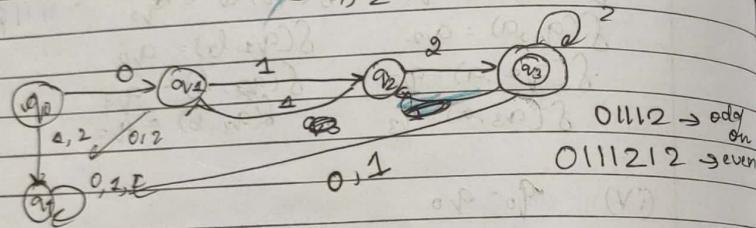
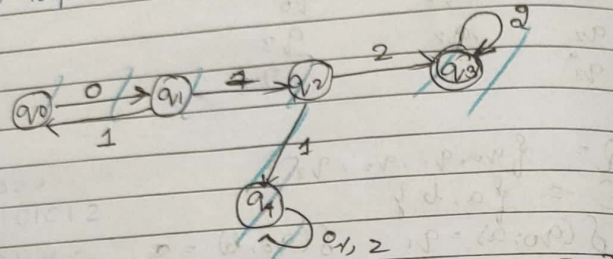
(iv)  $q_0 = q_0$

(v)  $F = \{q_3\}$

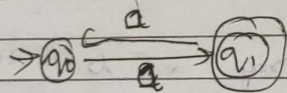
Draw DFA string a & b don't end with aab



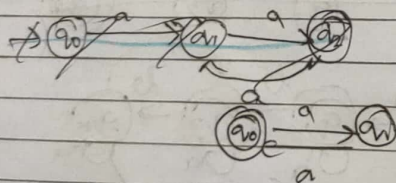
Design a DFA to accept string 0's, 1's, 2's beginning a '0' followed by odd number of 1's & ending with 2. Draw trap state also.



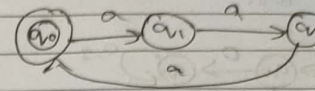
Design a DFA to accept odd no' of 'a'.



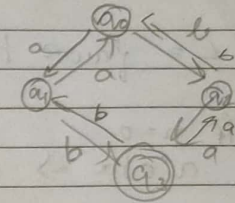
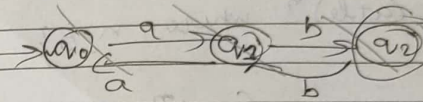
for even



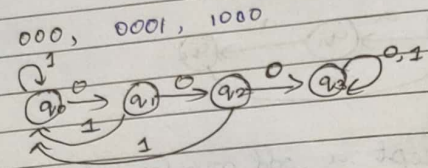
1.  $\{w \mid |w| \bmod 3 = 0\} \Sigma(a)$



Design a DFA to accept a odd number of a's and odd no' of b's



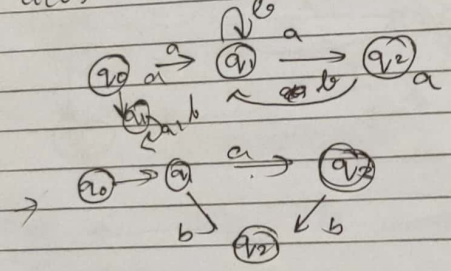
Draw DFA of string 0, 1 consecutive three



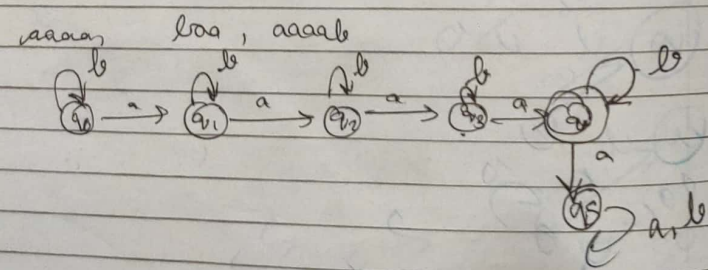
$L = \{aw^n \mid w \in (a+b)^n \text{ where } n \geq 0\}$

if  $n=0$

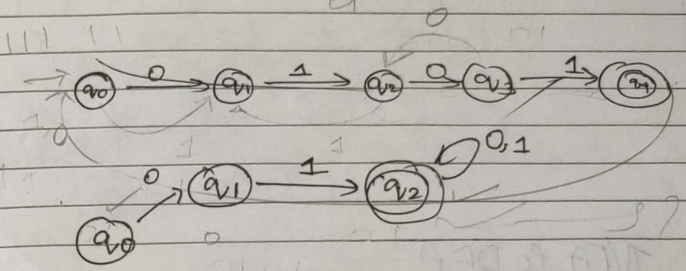
$a(a+b)^0 \Rightarrow aa$



DFA to accept a string a & b having 4 'a' where  $\Sigma = a, b$



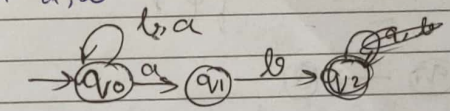
DFA set of all string of all alphabet  $\Sigma = 0, 1$ , end either begin and or end both with 01



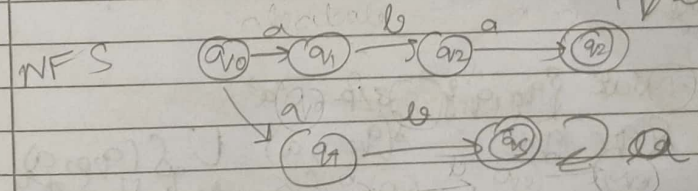
NFA:

$M = (Q, \Sigma, \delta, q_0, F)$

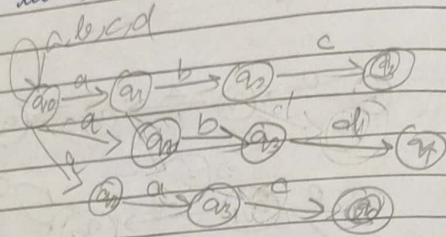
Design a NFA to accept string of a's & b's ending with a, b



$L = \{w \mid w \in abab^n \text{ or } abab^n \mid \text{where } n \geq 0\}$



NFA string  
(i) abc, abd, aacd



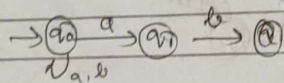
NFA to DFA

→ Subset Method

→ Lazy Evaluation Method

ex.  $\{q_0, q_1, q_2\}$   $\Sigma = \{a, b\}$

$\{q_0\}, \{q_1\}, \{q_2\}, \{q_1, q_0\}, \{q_1, q_2\}, \{q_0, q_2\},$   
 $\{q_0, q_1, q_2\}, \{\emptyset\}$



② Start state is same for both.

③ Identify FOS in subset = 4

④ Table

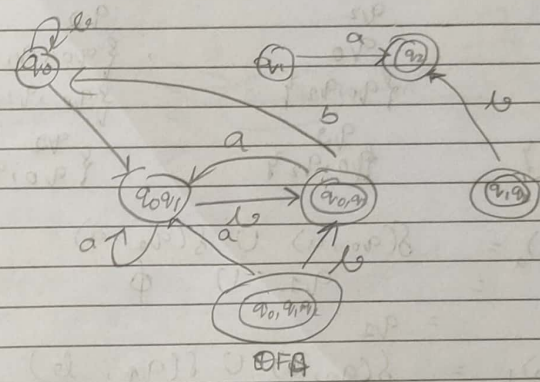
computation table

for state  $\{q_0, q_1\}$  I/P 'a'

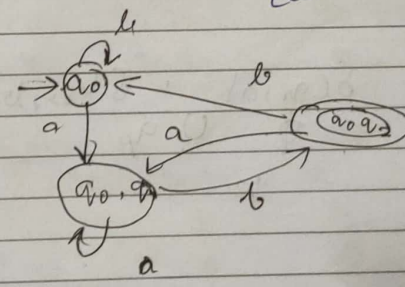
$$\delta(q_0, q_1) = \delta(q_0, a) \cup \delta(q_1, a) \\ = \{q_0, q_1\} \cup \{\emptyset\} \\ = \{q_0, q_1\}$$

$$\{q_1, q_2\} \xrightarrow{a} \{q_1\} \cup \{q_2\} \xrightarrow{b} \{q_2\}$$

|                     | a              | b              |
|---------------------|----------------|----------------|
| $\{q_0\}$           | $\{q_0, q_1\}$ | $\{q_0\}$      |
| $\{q_1\}$           | $\emptyset$    | $\{q_2\}$      |
| $\{q_2\}$           | $\emptyset$    | $\emptyset$    |
| $\{q_0, q_1\}$      | $\{q_0, q_1\}$ | $\{q_0, q_2\}$ |
| $\{q_0, q_2\}$      | $\{q_0, q_1\}$ | $\{q_0, q_2\}$ |
| $\{q_1, q_2\}$      | $\emptyset$    | $\{q_2\}$      |
| $\{q_0, q_1, q_2\}$ | $\{q_0, q_1\}$ | $\{q_0, q_2\}$ |



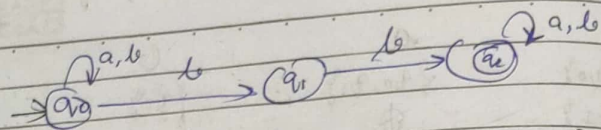
Note: As  $\{q_0\}, \{q_2\}, \{q_1, q_2\}$  &  $\{q_0, q_1, q_2\}$  are not reachable from start state we can remove it



$M = \{Q, \Sigma, \delta, q_0, F\}$   
 $Q = \{q_0, q_1\}$   
 $\{q_0, q_1\}$

$\Sigma = \{a, b\}$

$q_0 = q_0$   
 $F = \{q_1\}$



$$\delta(q_0, q_1, q_2) = \emptyset, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}$$

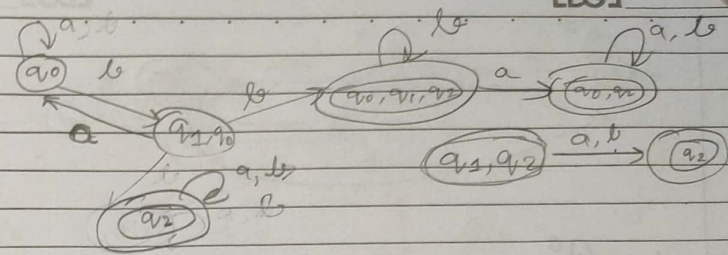
| Q/E                                                   | a                                  | b                                                   |
|-------------------------------------------------------|------------------------------------|-----------------------------------------------------|
| → q <sub>0</sub>                                      | q <sub>0</sub>                     | {q <sub>0</sub> , q <sub>1</sub> }                  |
| q <sub>1</sub>                                        | ∅                                  | {q <sub>2</sub> }                                   |
| * q <sub>2</sub>                                      | q <sub>2</sub>                     | q <sub>2</sub>                                      |
| * {q <sub>0</sub> , q <sub>1</sub> }                  | q <sub>0</sub>                     | {q <sub>0</sub> , q <sub>1</sub> , q <sub>2</sub> } |
| * {q <sub>0</sub> , q <sub>2</sub> }                  | {q <sub>0</sub> , q <sub>2</sub> } | {q <sub>0</sub> , q <sub>2</sub> }                  |
| * {q <sub>1</sub> , q <sub>2</sub> }                  | q <sub>2</sub>                     | q <sub>2</sub>                                      |
| * {q <sub>0</sub> , q <sub>1</sub> , q <sub>2</sub> } | {q <sub>0</sub> , q <sub>2</sub> } | {q <sub>0</sub> , q <sub>1</sub> , q <sub>2</sub> } |

$$\begin{aligned} \delta(q_0, q_1)_a &= \delta(q_0, a) \cup \delta(q_1, a) \\ &= q_1 \cup \emptyset \\ &= q_1 \end{aligned}$$

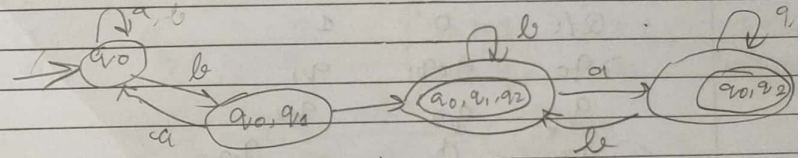
$$\begin{aligned} \delta(q_0, q_2)_b &= \delta(q_0, b) \cup \delta(q_2, b) \\ &= \{q_0, q_1\} \cup \{q_2\} \\ &= \{q_0, q_1, q_2\} \end{aligned}$$

||ly {q<sub>0</sub>, q<sub>2</sub>} {q<sub>1</sub>, q<sub>2</sub>} {q<sub>1</sub>, q<sub>0</sub>, q<sub>2</sub>} are derived.

$$\begin{aligned} \delta(q_1, q_2) &= \delta(q_1, a) + \delta(q_2, b) \\ &= \emptyset \cup q_2 \\ &= q_2 \end{aligned}$$

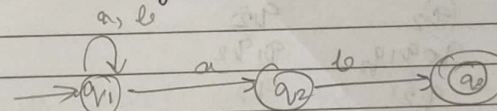


From start state q<sub>2</sub>, {q<sub>1</sub>, q<sub>2</sub>}, q<sub>2</sub> doesn't occur so the final deduce form is



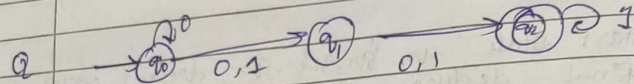
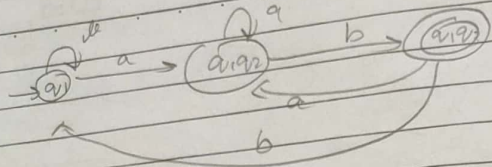
Lazy Method

Q NFA ending with ab



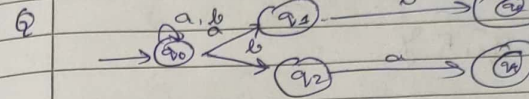
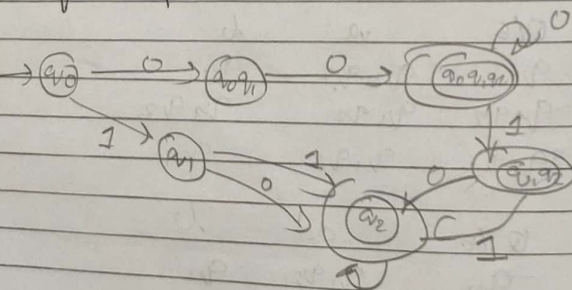
| Q/E                             | a                             | b                             |
|---------------------------------|-------------------------------|-------------------------------|
| → q <sub>1</sub>                | q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub>                |
| q <sub>2</sub> q <sub>3</sub>   | q <sub>1</sub> q <sub>2</sub> | q <sub>2</sub> q <sub>3</sub> |
| * q <sub>1</sub> q <sub>3</sub> | q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub>                |

| Q/E                             | a                             | b                             |
|---------------------------------|-------------------------------|-------------------------------|
| → q <sub>1</sub>                | q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub>                |
| q <sub>2</sub>                  | ∅                             | q <sub>1</sub> q <sub>3</sub> |
| * q <sub>1</sub> q <sub>3</sub> | ∅                             | ∅                             |



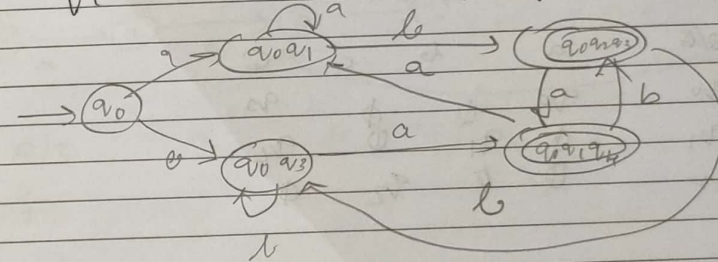
| Q/E               | 0           | 1     |
|-------------------|-------------|-------|
| $\rightarrow q_0$ | $q_0 q_1$   | $q_1$ |
| $q_1$             | $q_2$       | $q_2$ |
| $q_2$             | $\emptyset$ | $q_2$ |

| Q/E           | 0             | 1         |
|---------------|---------------|-----------|
| $q_0$         | $q_0 q_1$     | $q_1$     |
| $q_0 q_1$     | $q_0 q_1 q_2$ | $q_1 q_2$ |
| $q_1$         | $q_2$         | $q_2$     |
| $q_0 q_1 q_2$ | $q_0 q_1 q_2$ | $q_1 q_2$ |
| $q_1 q_2$     | $q_2$         | $q_2$     |
| $q_2$         | $\emptyset$   | $q_2$     |



| Q/E               | a           | b           |
|-------------------|-------------|-------------|
| $\rightarrow q_0$ | $q_0 q_1$   | $q_0 q_2$   |
| $q_1$             | $\emptyset$ | $q_2$       |
| $q_2$             | $\emptyset$ | $\emptyset$ |
| $q_3$             | $q_1$       | $\emptyset$ |
| $q_4$             | $\emptyset$ | $\emptyset$ |

| Q/E               | a             | b             |
|-------------------|---------------|---------------|
| $\rightarrow q_0$ | $q_0 q_1$     | $q_0 q_2$     |
| $q_0 q_1$         | $q_0 q_1$     | $q_0 q_1 q_2$ |
| $q_0 q_2$         | $q_0 q_1 q_2$ | $q_1 q_2$     |
| $q_0 q_1 q_2$     | $q_0 q_1 q_2$ | $q_0 q_2$     |
| $q_0 q_1 q_2$     | $q_0 q_1$     | $q_0 q_2$     |



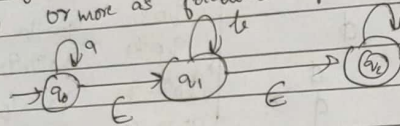
$q_0$  is start state as it is the start state  
NFA

$q_0 q_2 q_3$ ,  $q_0 q_1 q_2$  are final states  
 $q_2$  &  $q_4$  are the respective final states

## E-NFA:-

$$M = \{Q, \Sigma, \delta, q_0, F\}$$

Ex: design E-NFA which accept string accepting 0 or more a followed by 0 or more b followed by 0 or more c



| Q/E            | a              | b              | c              | ε              |
|----------------|----------------|----------------|----------------|----------------|
| q <sub>0</sub> | q <sub>0</sub> | ∅              | ∅              | q <sub>1</sub> |
| q <sub>1</sub> | ∅              | q <sub>1</sub> | ∅              | q <sub>2</sub> |
| q <sub>2</sub> | ∅              | ∅              | q <sub>2</sub> | ∅              |

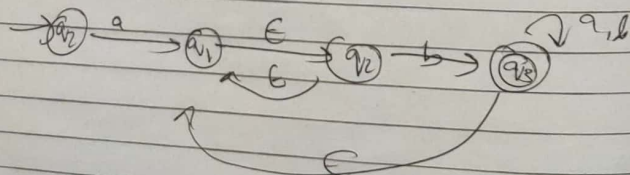
## ε closure

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

Ex:

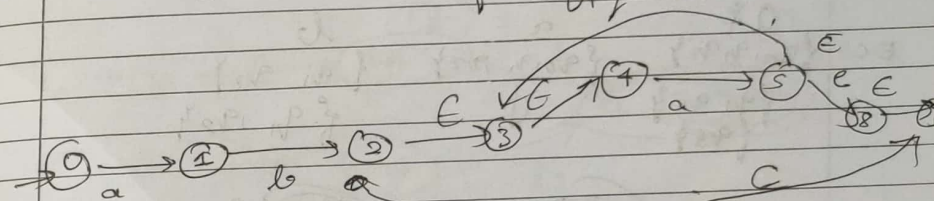


$$\epsilon\text{-closure}(q_0) = \{q_0\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2, q_1\}$$

$$\epsilon\text{-closure}(q_3) = \{q_3, q_1\}$$



| Q/E            | a              | b              | c              | ε              |
|----------------|----------------|----------------|----------------|----------------|
| q <sub>0</sub> | q <sub>1</sub> | ∅              | ∅              | ∅              |
| q <sub>1</sub> | q <sub>1</sub> | q <sub>2</sub> | ∅              | q <sub>4</sub> |
| q <sub>2</sub> | ∅              | q <sub>2</sub> | q <sub>3</sub> | ∅              |
| q <sub>3</sub> | ∅              | ∅              | q <sub>3</sub> | q <sub>4</sub> |
| q <sub>4</sub> | ∅              | ∅              | ∅              | q <sub>1</sub> |

$$\epsilon\text{-closure}(q_0) = \{q_0\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2, q_3, q_4\}$$

$$\epsilon\text{-closure}(q_3) = \{q_3, q_4, q_2\}$$

$$\epsilon\text{-closure}(q_4) = \{q_4, q_1\}$$

$$\epsilon\text{-closure}(q_5) = \{q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30}, q_{31}, q_{32}, q_{33}, q_{34}, q_{35}, q_{36}, q_{37}, q_{38}, q_{39}, q_{40}, q_{41}, q_{42}, q_{43}, q_{44}, q_{45}, q_{46}, q_{47}, q_{48}, q_{49}, q_{50}, q_{51}, q_{52}, q_{53}, q_{54}, q_{55}, q_{56}, q_{57}, q_{58}, q_{59}, q_{60}, q_{61}, q_{62}, q_{63}, q_{64}, q_{65}, q_{66}, q_{67}, q_{68}, q_{69}, q_{70}, q_{71}, q_{72}, q_{73}, q_{74}, q_{75}, q_{76}, q_{77}, q_{78}, q_{79}, q_{80}, q_{81}, q_{82}, q_{83}, q_{84}, q_{85}, q_{86}, q_{87}, q_{88}, q_{89}, q_{90}, q_{91}, q_{92}, q_{93}, q_{94}, q_{95}, q_{96}, q_{97}, q_{98}, q_{99}\}$$

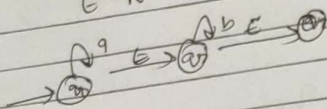
$$\epsilon\text{-closure}(q_6) = \{q_6\}$$

$$\epsilon\text{-closure}(q_7) = \{q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30}, q_{31}, q_{32}, q_{33}, q_{34}, q_{35}, q_{36}, q_{37}, q_{38}, q_{39}, q_{40}, q_{41}, q_{42}, q_{43}, q_{44}, q_{45}, q_{46}, q_{47}, q_{48}, q_{49}, q_{50}, q_{51}, q_{52}, q_{53}, q_{54}, q_{55}, q_{56}, q_{57}, q_{58}, q_{59}, q_{60}, q_{61}, q_{62}, q_{63}, q_{64}, q_{65}, q_{66}, q_{67}, q_{68}, q_{69}, q_{70}, q_{71}, q_{72}, q_{73}, q_{74}, q_{75}, q_{76}, q_{77}, q_{78}, q_{79}, q_{80}, q_{81}, q_{82}, q_{83}, q_{84}, q_{85}, q_{86}, q_{87}, q_{88}, q_{89}, q_{90}, q_{91}, q_{92}, q_{93}, q_{94}, q_{95}, q_{96}, q_{97}, q_{98}, q_{99}\}$$

$$\epsilon\text{-closure}(q_8) = \{q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30}, q_{31}, q_{32}, q_{33}, q_{34}, q_{35}, q_{36}, q_{37}, q_{38}, q_{39}, q_{40}, q_{41}, q_{42}, q_{43}, q_{44}, q_{45}, q_{46}, q_{47}, q_{48}, q_{49}, q_{50}, q_{51}, q_{52}, q_{53}, q_{54}, q_{55}, q_{56}, q_{57}, q_{58}, q_{59}, q_{60}, q_{61}, q_{62}, q_{63}, q_{64}, q_{65}, q_{66}, q_{67}, q_{68}, q_{69}, q_{70}, q_{71}, q_{72}, q_{73}, q_{74}, q_{75}, q_{76}, q_{77}, q_{78}, q_{79}, q_{80}, q_{81}, q_{82}, q_{83}, q_{84}, q_{85}, q_{86}, q_{87}, q_{88}, q_{89}, q_{90}, q_{91}, q_{92}, q_{93}, q_{94}, q_{95}, q_{96}, q_{97}, q_{98}, q_{99}\}$$

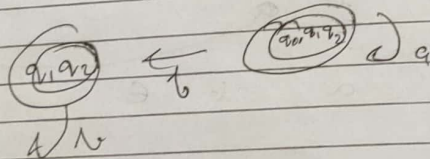
$$\epsilon\text{-closure}(q_9) = \{q_9\}$$

E-NFA to DFA

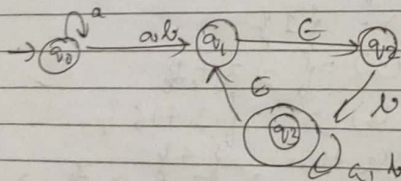


E- $\delta$  for  $q_0 = \{q_0, q_1, q_2\}$   
 $E\delta$   $q_1 = \{q_1, q_2\}$   
 $E\delta$   $q_2 = \{q_2\}$

|       | a                   | b              |
|-------|---------------------|----------------|
| $q_0$ | $\{q_0, q_1, q_2\}$ | $\{q_1, q_2\}$ |
| $q_1$ | $\emptyset$         | $\{q_1, q_2\}$ |
| $q_2$ | $\emptyset$         | $\emptyset$    |

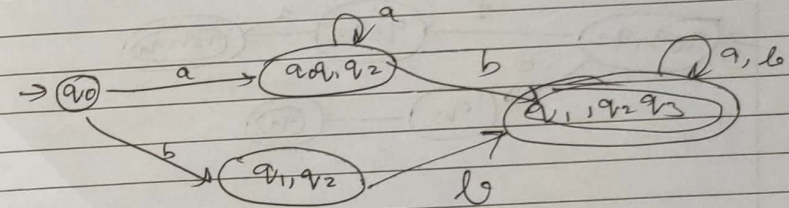


E-NFA to

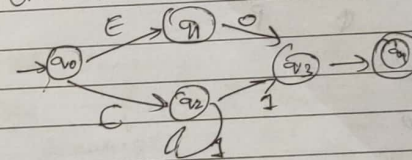


$E\delta(q_0) = \{q_0\}$   
 $E\delta(q_1) = \{q_1, q_2\}$   
 $E\delta(q_2) = \{q_2\}$   
 $E\delta(q_3) = \{q_3, q_1, q_2\}$

|       | a                   | b                   |
|-------|---------------------|---------------------|
| $q_0$ | $\{q_0, q_1, q_2\}$ | $\{q_1, q_2\}$      |
| $q_1$ | $\{q_0, q_1, q_2\}$ | $\{q_1, q_2, q_3\}$ |
| $q_2$ | $\emptyset$         | $\{q_1, q_2, q_3\}$ |
| $q_3$ | $\{q_1, q_2, q_3\}$ | $\{q_1, q_2, q_3\}$ |



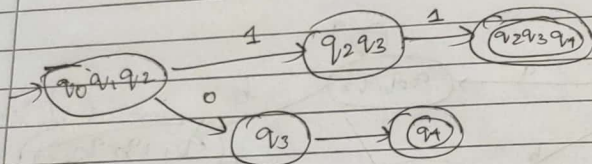
ENFA to DFA



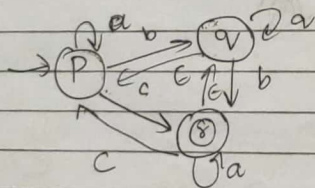
$E\delta(q_0) = \{q_0, q_1, q_2\}$   
 $E\delta(q_1) = \{q_1\}$

|       | a                   | b              |
|-------|---------------------|----------------|
| $q_0$ | $\{q_0, q_1, q_2\}$ | $\{q_1, q_2\}$ |
| $q_1$ | $\{q_1\}$           | $\emptyset$    |
| $q_2$ | $\emptyset$         | $\{q_1, q_2\}$ |
| $q_3$ | $\emptyset$         | $\emptyset$    |

| Q/E                       | 0           | 1             |
|---------------------------|-------------|---------------|
| $\rightarrow q_0 q_1 q_2$ | $q_3$       | $q_2 q_3$     |
| $q_3$                     | $\emptyset$ | $q_1$         |
| $q_2 q_3$                 | $\emptyset$ | $q_2 q_3 q_1$ |
| $q_2 q_3 q_1$             | $\emptyset$ | $q_2 q_3 q_1$ |
| $q_1$                     | $\emptyset$ | $\emptyset$   |



| Q:              | $\epsilon$  | a | b           | c           |
|-----------------|-------------|---|-------------|-------------|
| $\rightarrow p$ | $\emptyset$ | p | q           | r           |
| q               | p           | q | r           | $\emptyset$ |
| * r             | q           | r | $\emptyset$ | p           |

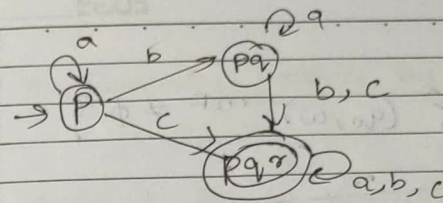


$$\epsilon - C = \{p\}$$

$$\epsilon C = \{q, p\}$$

$$\epsilon C r = \{r, q, p\}$$

|                 | a   | b   | c   |
|-----------------|-----|-----|-----|
| $\rightarrow p$ | p   | pq  | pqr |
| * pq            | pq  | pqr | pqr |
| pqr             | pqr | pqr | pqr |



### Theory

Language of NFA, DFA

Subset theorem

Application of FA

### Language of DFA

$$DFA = A = (Q, \Sigma, \delta, q_0, F)$$

$$L(A) = \{w \mid \delta(q_0, w) \text{ is in } F\}$$

This language is denoted by  $L(A)$  & is defined by  $\{w \mid \delta(q_0, w) \text{ is in } F\}$ .

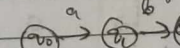
A language set of strings  $w$ , with take  $q_0$  is start states to the one of the final state, if  $L(A)$  for some DFA, then we say 'L' is a regular language.

Transition Function

Extended T.F

$$\delta(q_0, a) = q_1$$

$$w = a, b$$



## Language of NFA

$$L(A) = \{ w \mid \delta(q_0, w), w \neq \emptyset \}$$

Regular Expression  $^*$ ,  $+$ ,  $^?$   
closure concat Union

Regular language: NFA, DFA, ENFA accepted by  $\Gamma$

Regular Language: is a language accepted by DFA, NFA, ENFA can be designed using regular expression consisting of symbols such as  $-, +, ^*$ .

$+$  - Union

$*$  - Closure

$\cdot$  - concatenation

Recursively defined as

$\emptyset$  is a RE denoting empty string

$\epsilon$  is a RE indicating lang contain empty string

$a$  is a RE indicating lang contain only  $\{a\}$

If  $R$  is RE denoting lang  $\{R\}$  &  $S$  is RE denoting lang  $\{S\}$

$R+S$  is RE corresponding to

$L_{R+L_S}$

$R \cdot S$  is RE corresponding to  $L_R \cdot L_S$

$R^*$  is RE corresponding to  $L_R^*$

The expression obtain by above rules are called RE

4) Design a Regular Expression (RE) containing string any numbers of string  $a$

$a^*$

Any no  $a$   $RE = a^*$   
atleast one  $a$   $RE = aa^*$   
only  $a$   $a^+$

Either one  $a$  or one  $b$   $RE = (a+b)$

Set of a string  $a$ 's or  $b$ 's of any length with  $\emptyset$

$$RE = (a+b)^*$$

Set of string  $a$  &  $b$  ending  $ab$   
 $RE = (a+b)^* \cdot ab$

Set of string  $a$  &  $b$  having sub string  $aa$   
 $RE = aa^+(a+b)^* \cdot ab$

Set of string  $a$  &  $b$  either  $a$  or  $bb$   
 $RE = (a+b)^* \cdot (a+bb)$

\* even no a's

$$(b)^*$$

\* whose length is  $2 \in (a, b)$

$$(a+b) \cdot (a+b)$$

\* whose length  $\leq 2 \in (a, b)$

$$(a+b) + (a+b) \quad (E + a + b)$$

\* having even length

$$((a+b) \cdot (a+b))^*$$

\* odd length

$$(a+b) + ((a+b) \cdot (a+b))^*$$

\* 3 consecutive a's

$$(a+b)^* a a a (a+b)^*$$

\* whose second symbol from right end is a

$$(a+b)^* a (a+b)$$

\* since that 3rd symbol from right is the 4th

$$(a+b)^* b a (a+b) (a+b)$$

\* whose length either even or multiple of 3 or both.

$$((a+b)(a+b))^* + ((a+b)(a+b)a(a+b))^*$$

Sender  $\rightarrow L = \{a^n b^m \mid m \text{ is even}\}$

case 1: even a + even b = even

Receiver: case 2: odd a + odd b = even  $\Rightarrow$  even

$$R = a(aa)^* b(bb)^* + (aa)^* (bb)^*$$

17)  $\Sigma = \{a, b\}$  containing not more than 3 a's

$$\Rightarrow ab^* + b^* a + b^* + b^* a b + b^* a b^*$$

$$b^* a b^* a b^* + b^* a b^* a b^*$$

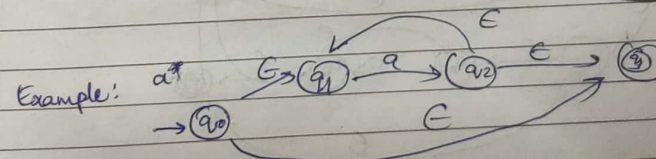
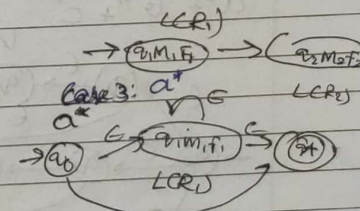
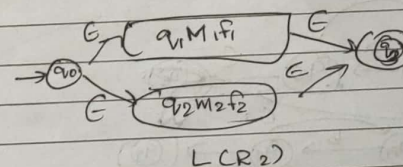
$$b^* (E + a) b^* (E + a) b^* (E + a) b^*$$

$(E + a) \Rightarrow$  mean E or a (E can be  $\phi$ )

RE to E-NFA

case 1:  $R = R_1 + R_2$

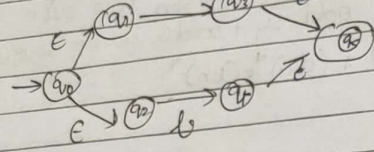
case 2:  $R = R_1 R_2$



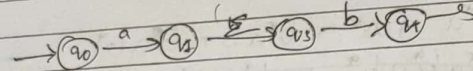
E. closure  $q_0 = q_0$

URBAN  
EDGE

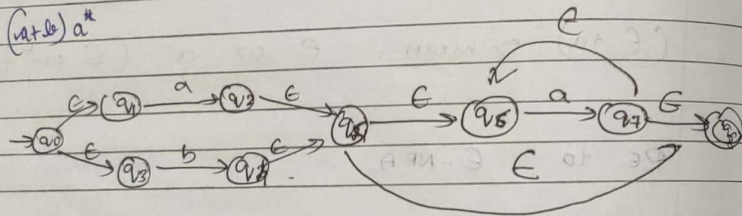
$a + b$



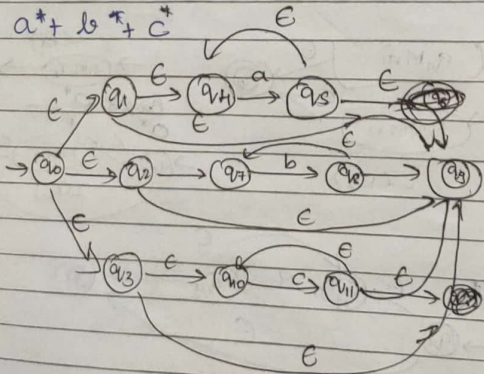
$ab$



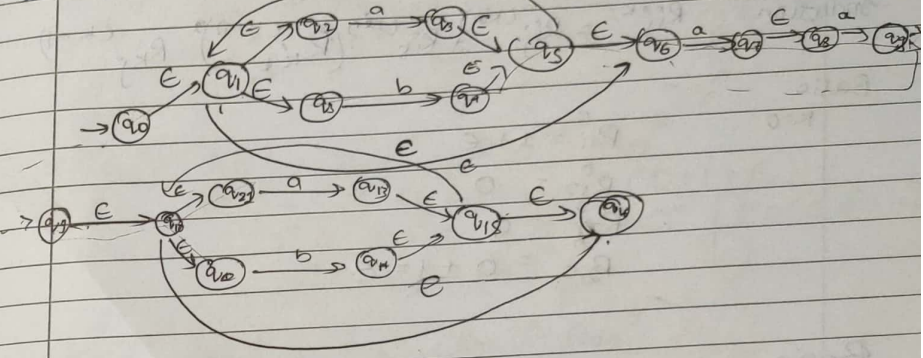
$(a+b)^*$



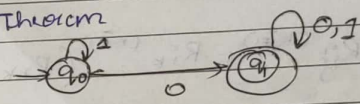
$a^* + b^* + c^*$



2)  $(a+tb)^* aa(a+b)^*$



Kleene's Theorem



Basic:  $R_{ij}^{(n)}$

Induction:  $R_{ij}^{(k)} = R_{ij}^{(k-1)} \cup R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$

$k \rightarrow$  No of state or state No

$i \rightarrow$  current state  $j \rightarrow$  Next state

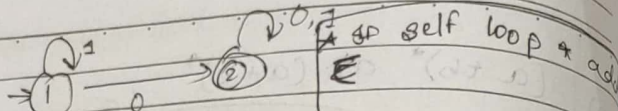
$R \rightarrow$  Regex

How many  $k =$  Total no of state FA

$k = 2$

Final RoE

$R_{12} = ?$

Basic  $R_{ij}^0$ 

Induction  $R_{ij}^{0,k} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$

Basic

 $k=0$ 

$$R_{11}^0 = 1 + \epsilon$$

$$R_{12}^0 = 0$$

$$R_{21}^0 = \emptyset$$

$$R_{22}^0 = 0 + 1 = \epsilon$$

Basic

when  $k=1, 2$  use

$$R_{ij}^{0,k} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$$

$$R_{11}^1 = R_{11}^0 + R_{11}^0 (R_{11}^0)^* R_{11}^0$$

$$\begin{aligned} &= (1+\epsilon) + (1+\epsilon)(1+\epsilon)^* (1+\epsilon) \\ &\text{Reduced to } = (1+\epsilon) + (1+\epsilon)1^* (1+\epsilon) = 1^* \end{aligned}$$

$$R_{12}^1 = R_{12}^0 + R_{11}^0 (R_{11}^0)^* R_{12}^0$$

$$= 0 + (1+\epsilon)(1+\epsilon)^* 0 = 0 + 1^* 0 = 1^* 0$$

$$R_{21}^1 = R_{21}^0 + R_{21}^0 (R_{21}^0)^* R_{21}^0$$

$$\begin{aligned} &= \emptyset + \emptyset (1+\epsilon)^* (1+\epsilon) \\ &= \emptyset + \emptyset 1^* = \emptyset \end{aligned}$$

$$R_{22}^1 = R_{22}^0 + R_{21}^0 (R_{11}^0)^* R_{12}^0$$

$$\begin{aligned} &= 0 + 1 + \epsilon + \emptyset (1+\epsilon)^* 0 \\ &= 0 + 1 + \epsilon \end{aligned}$$

 $R_{11}^2$ 

$$\begin{aligned} R_{11}^2 &= R_{11}^1 + R_{12}^1 (R_{22}^1)^* R_{21}^1 \\ &= 1^* + 1^* 0 (0 + 1 + \epsilon)^* \emptyset \\ &= 1^* \end{aligned}$$

$$R_{12}^2 = R_{12}^1 + R_{11}^1 (R_{11}^1)^* R_{12}^1$$

$$\begin{aligned} &= 1^* 0 + 1^* 0 (0 + 1 + \epsilon)^* (0 + 1 + \epsilon) \\ &= 1^* 0 + 1^* 0 (0 + 1 + \epsilon)^* [1^* 0 \epsilon + \epsilon + 1 + \epsilon] \\ &= 1^* 0 (0 + 1 + \epsilon)^* \end{aligned}$$

$$R_{21}^2 = R_{21}^1 + R_{22}^1 (R_{22}^1)^* R_{21}^1$$

$$\begin{aligned} &= \emptyset + (0 + 1 + \epsilon)(0 + 1 + \epsilon)^* \emptyset \\ &= \emptyset \end{aligned}$$

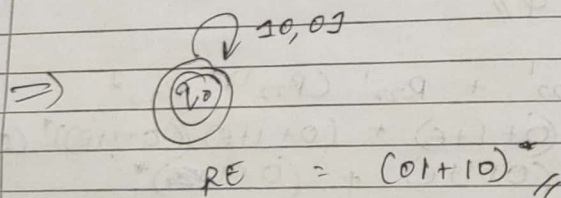
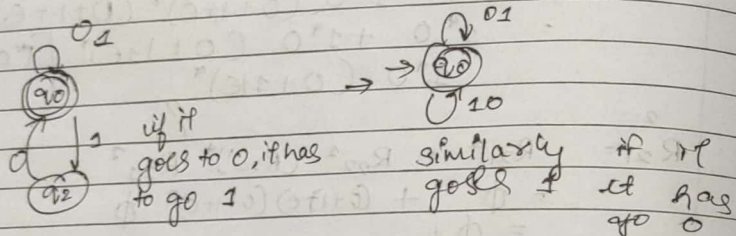
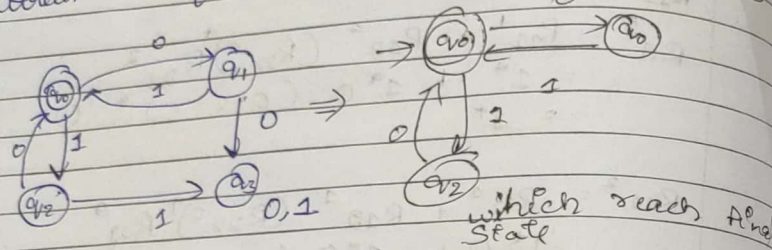
$$R_{22}^2 = R_{22}^1 + R_{21}^1 (R_{11}^1)^* R_{12}^1$$

$$\begin{aligned} &= (0 + 1 + \epsilon) + (0 + 1 + \epsilon)(0 + 1 + \epsilon)^* (0 + 1 + \epsilon) \\ &= (0 + 1 + \epsilon) + (0 + 1 + \epsilon)^* \end{aligned}$$

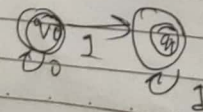
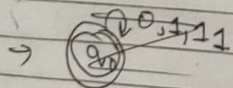
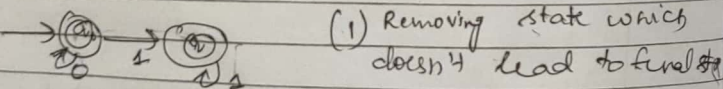
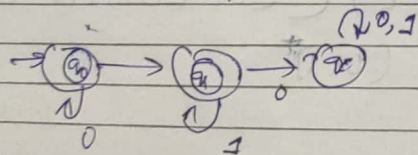
$$\begin{aligned} &= (0 + 1 + \epsilon) (\epsilon + (0 + 1 + \epsilon)^* (0 + 1 + \epsilon)) \\ &= (0 + 1 + \epsilon) \end{aligned}$$

Obtain RE from FA using Eliminating state

1



2

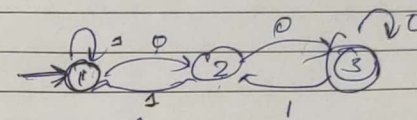


$$q_0 = RE = 0^*$$

$$q_1 RE = 0^* 1 1^*$$

$$= 0^* + 0^* 1 1^*$$

$$0^* (E + 11^*)$$



$$R_{11}^0 = 1 + E$$

$$R_{12}^0 = 0$$

$$R_{13}^0 = \emptyset \quad (\text{Not direct connecting})$$

$$R_{21}^0 = 1$$

$$R_{22}^0 = E$$

$$R_{23}^0 = 0$$

$$R_{31}^0 = \emptyset$$

$$R_{32}^0 = 1$$

$$R_{33}^0 = 0 + E$$

$$R_{ij}^k = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$$

$$R_{13}^3 = R_{13}^2 + R_{13}^2 (R_{33}^2)^* R_{33}^2$$

$$R_{13}^2 = R_{13}^0 + R$$

$$R_{11} = (1 + E) + (1 + E)^*$$

$$= (1 + E)^*$$

$$R_{12} = R_{12}^0 + (1 + E)^* 0$$

$$(1 + E)^* 0$$

